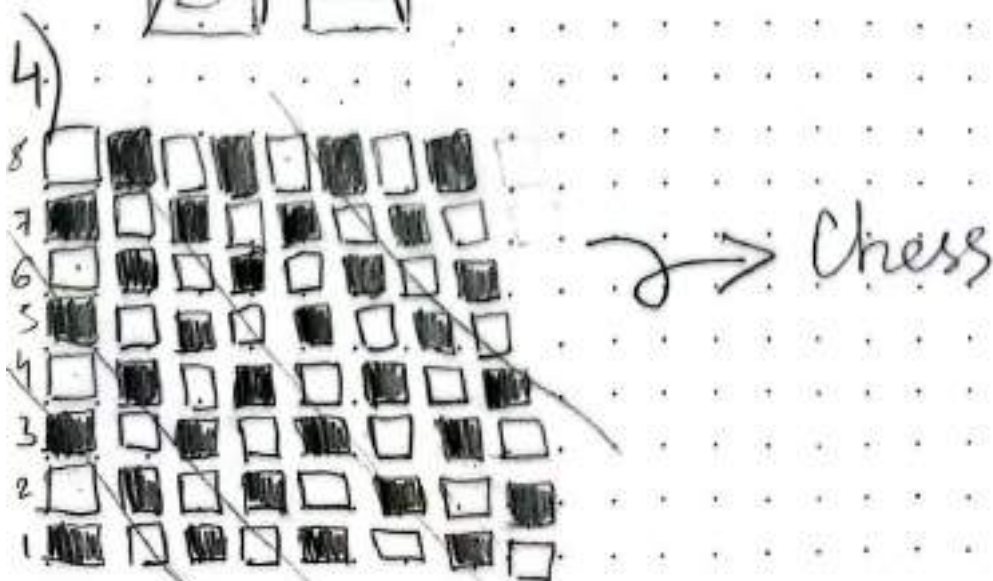
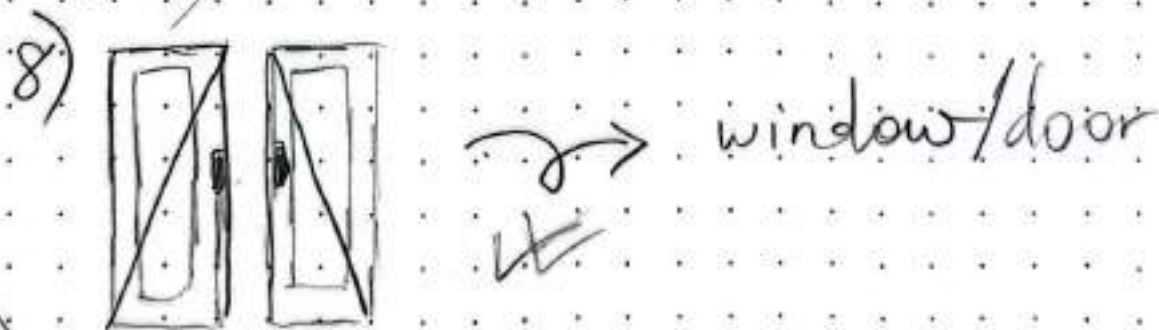
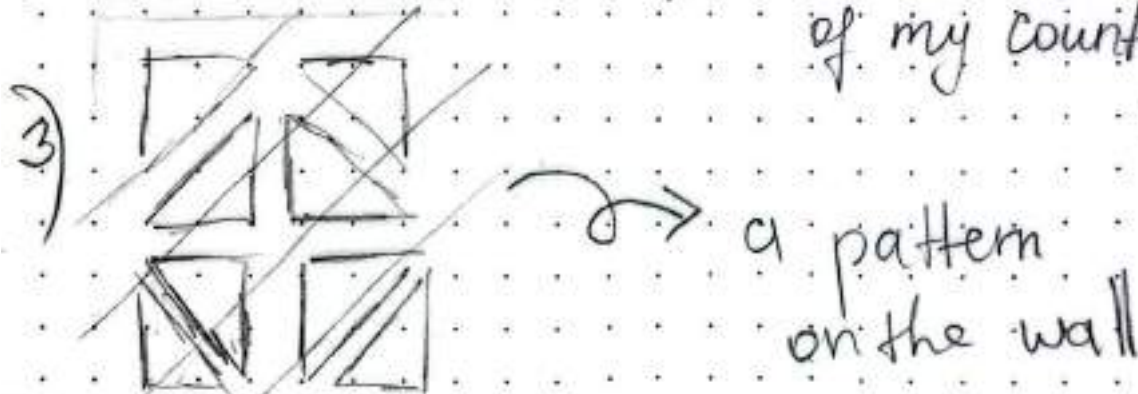
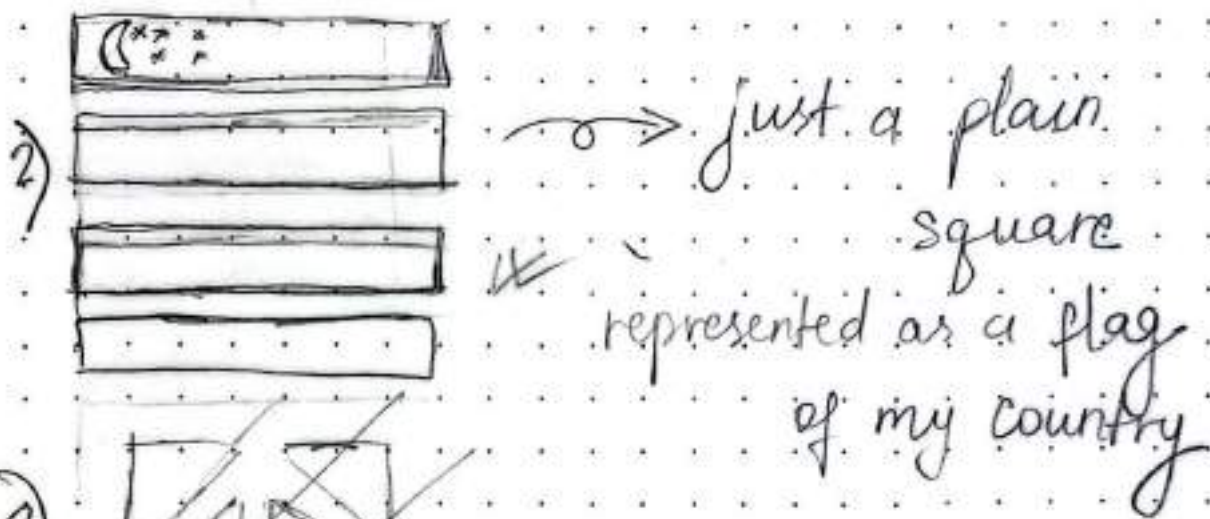
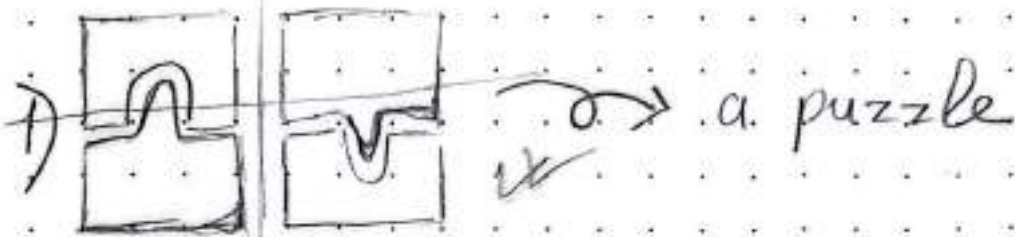
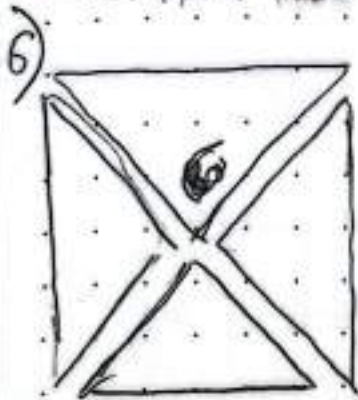


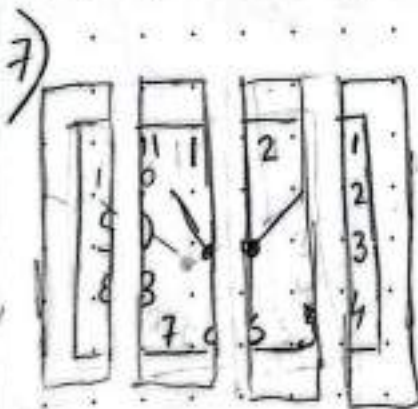




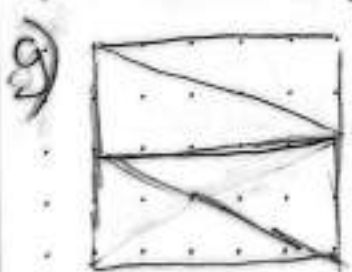
→ first aid
kit symbol



→ an envelope

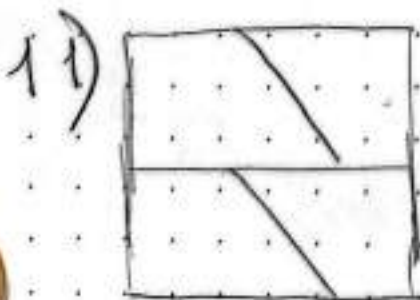
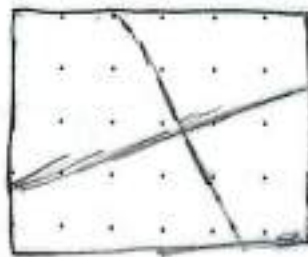


→ clock



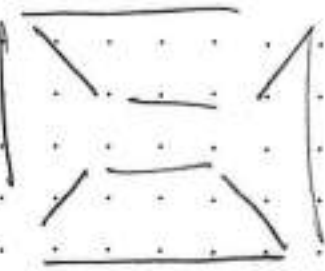
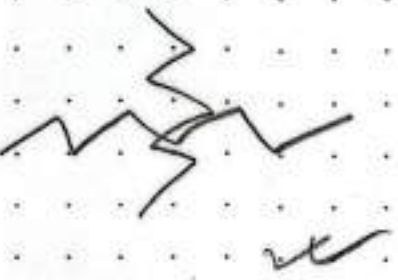
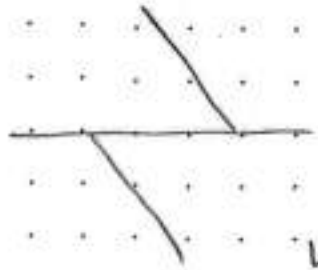
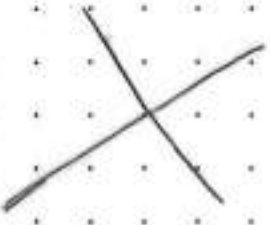
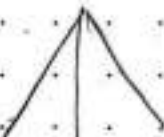
→ zigzag

(b)



→

can be
an image



Empathize → Define

Prototype ← Ideate

→ Test

Stenna: Creative Thinking with shapes standing up

↳ the next person:



I liked how Grace was accurate with cutting shapes.

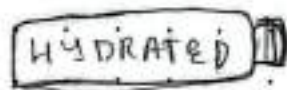


Oscar: Snow/winter mood!

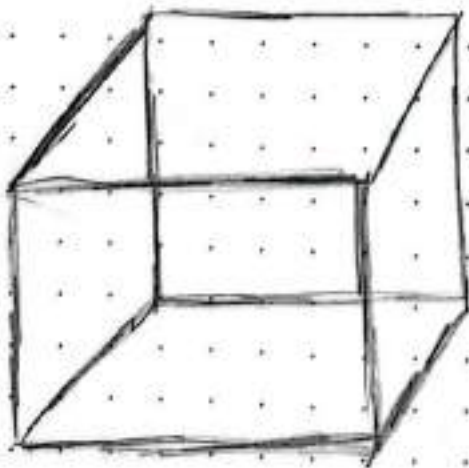
Paola: The illusional patterns.

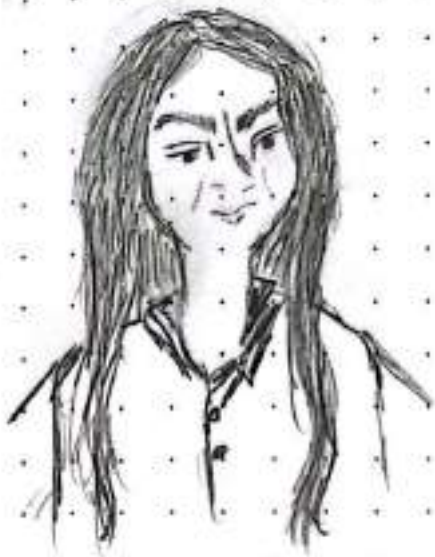
Go for quantity

Encouraging wild ideas

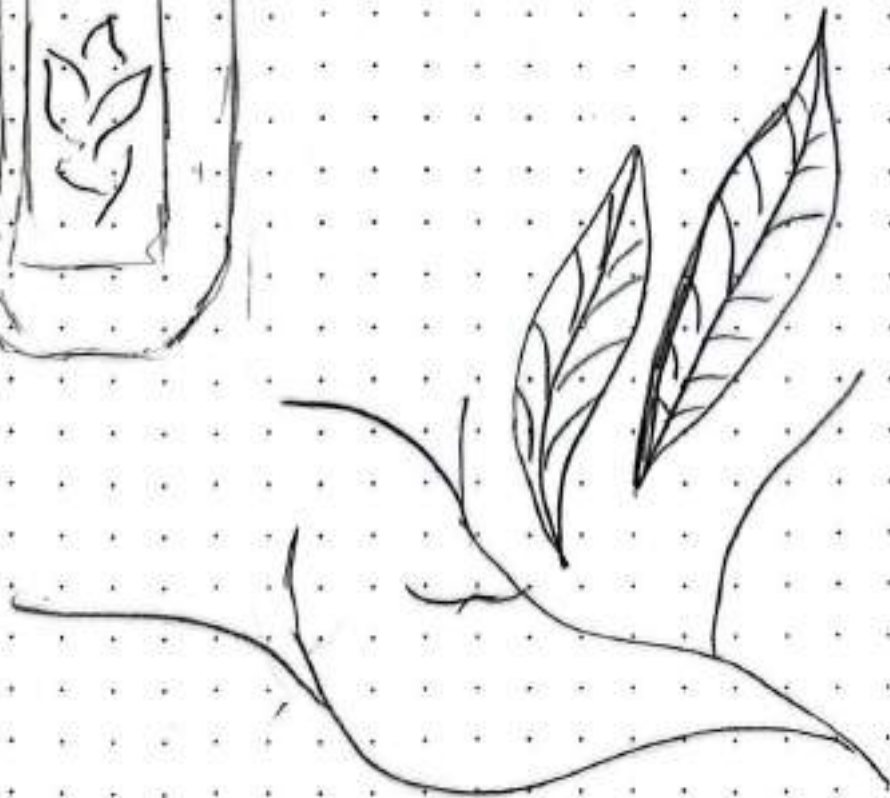


Late Night
Meal





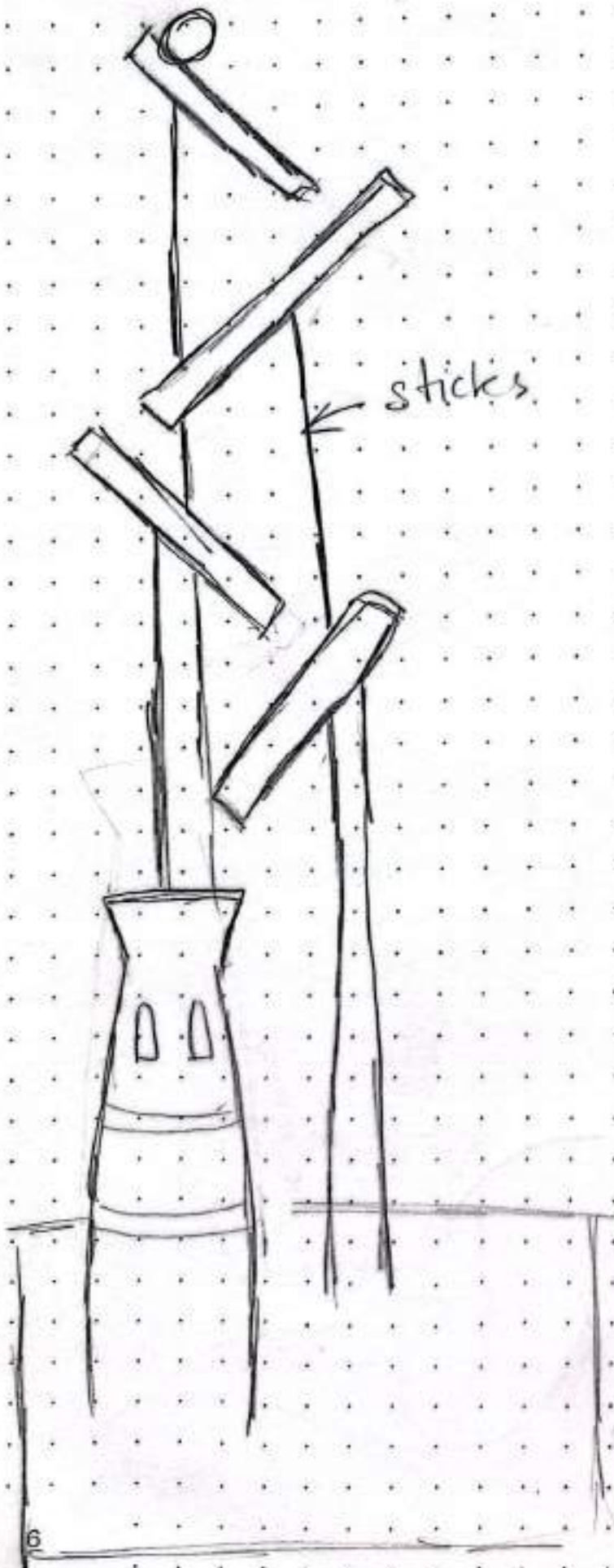
Jon!

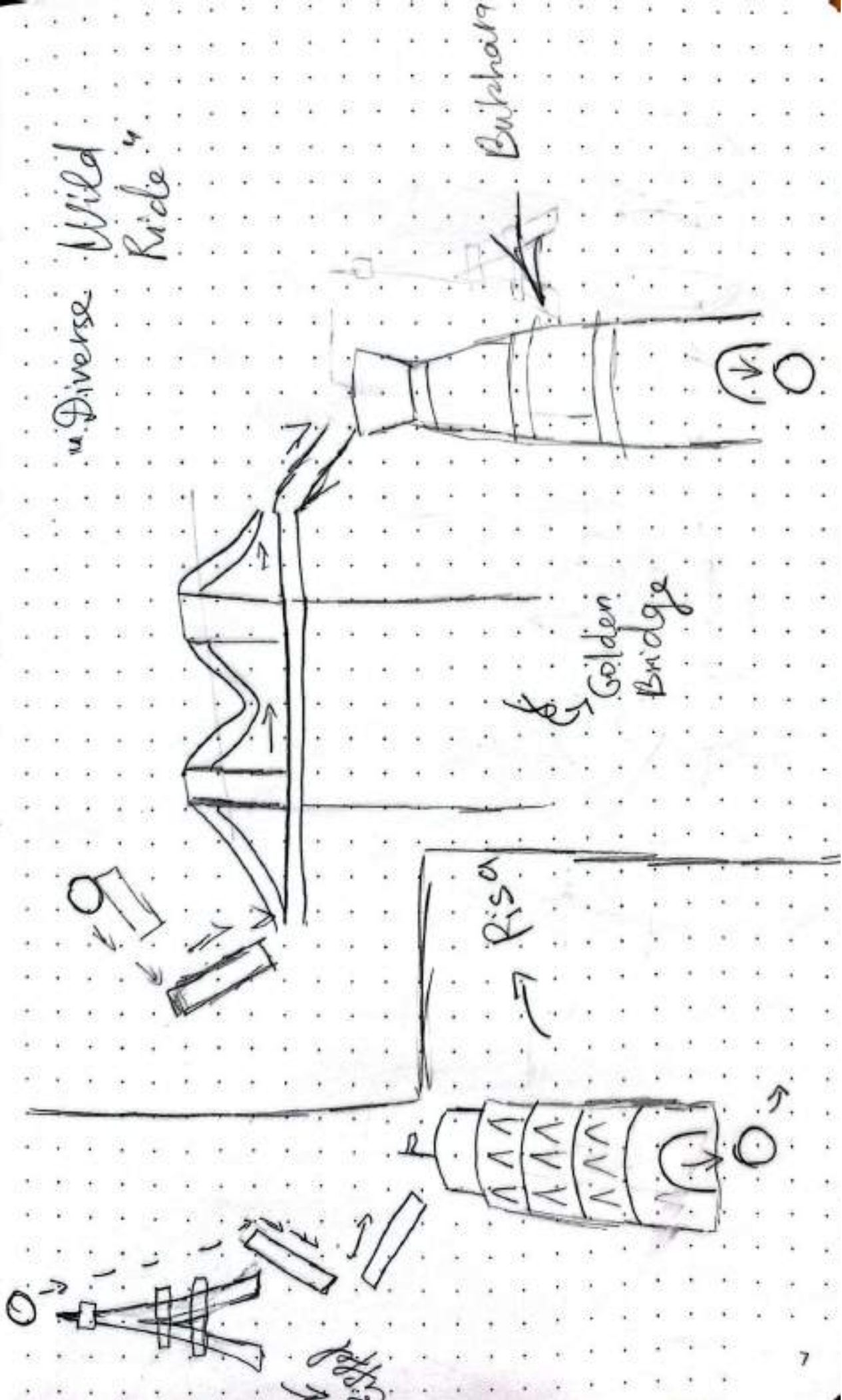


Project # 2

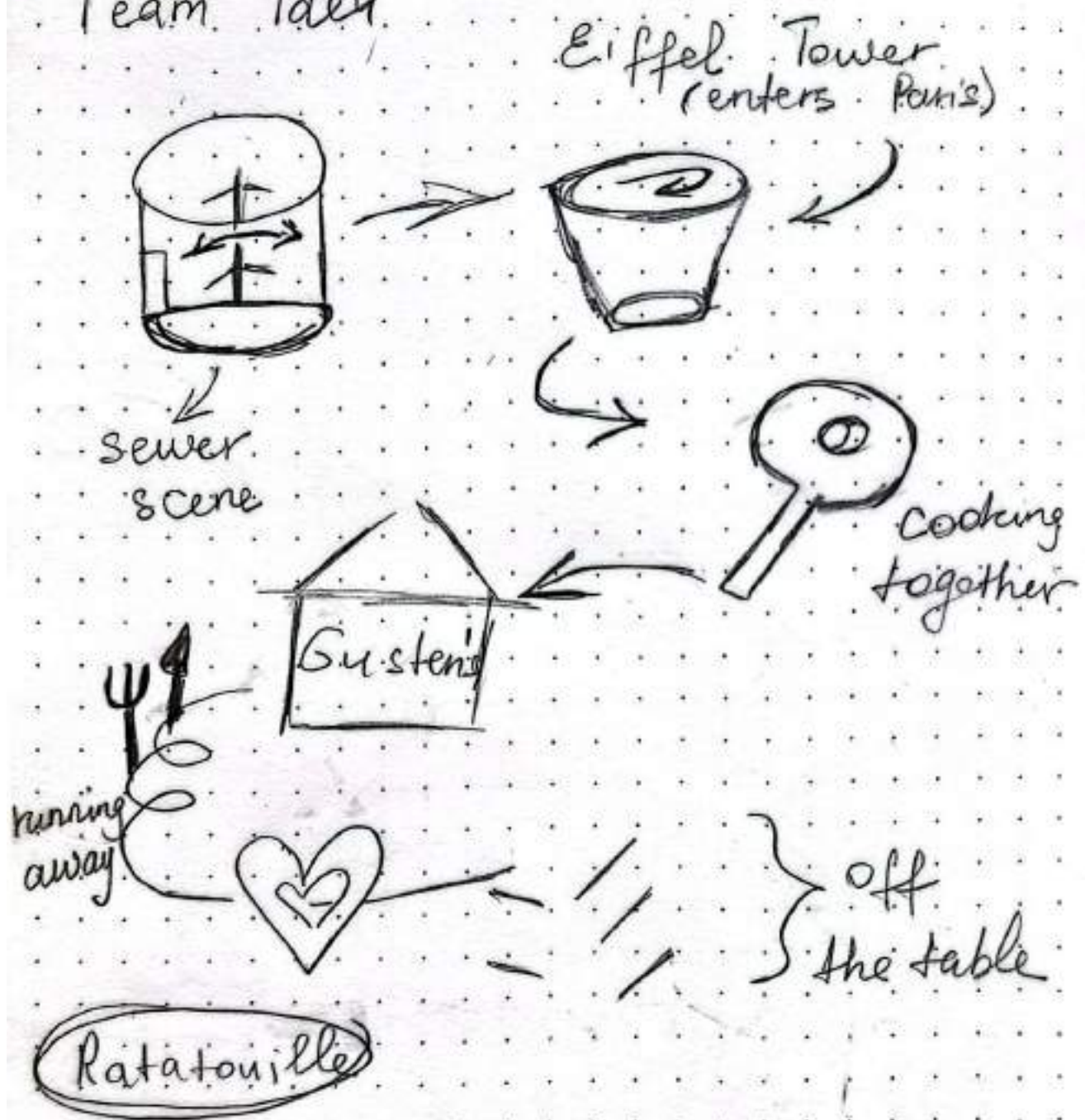
ball

sticks





Team Idea



Rat who wants to be a chef

Meets w/ Linguini

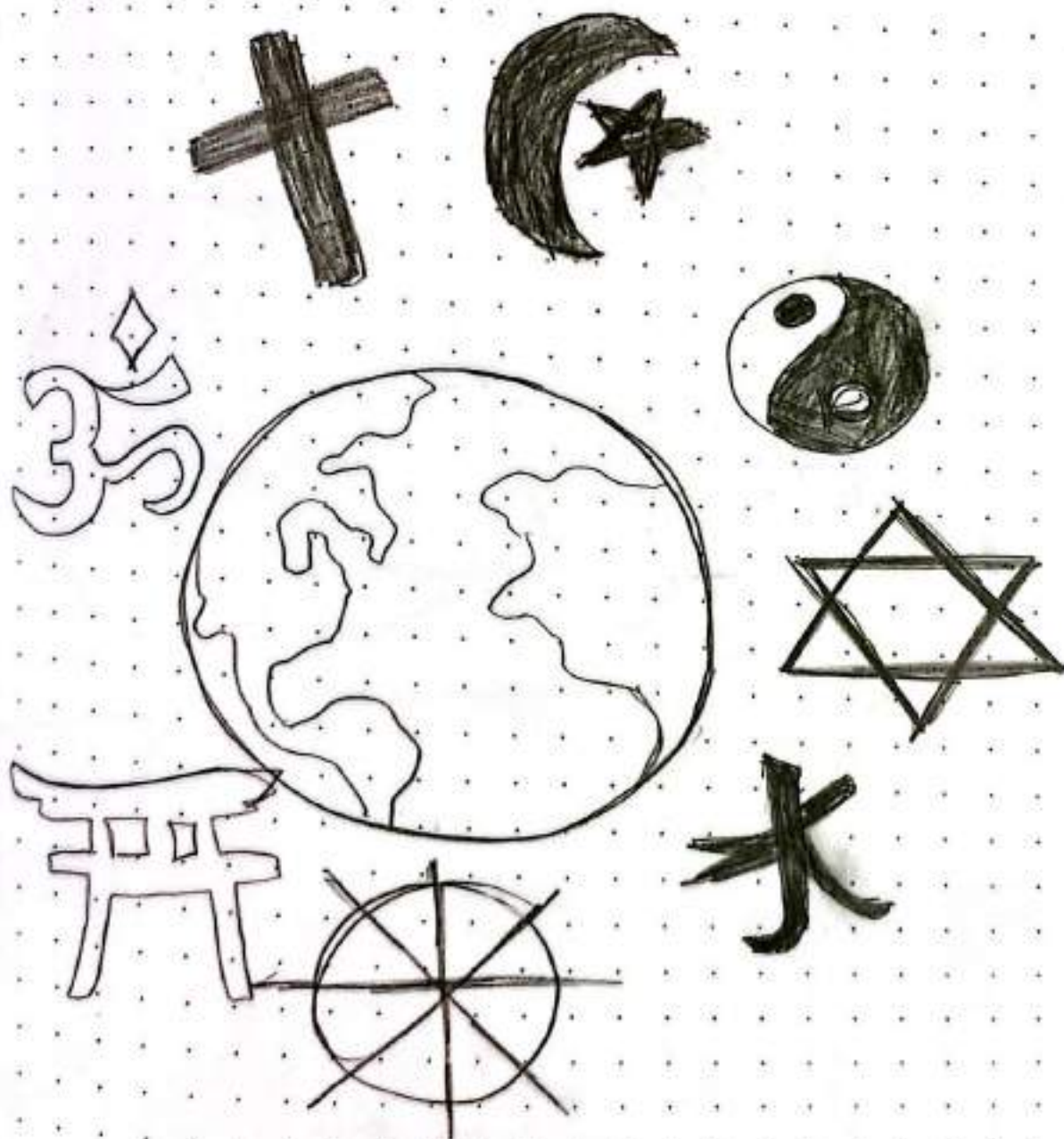
Linguini brings him to home

Rat helps Linguini w/ cooking

Linguini is found to be a son of Gusteau

Building a new restaurant

Delights

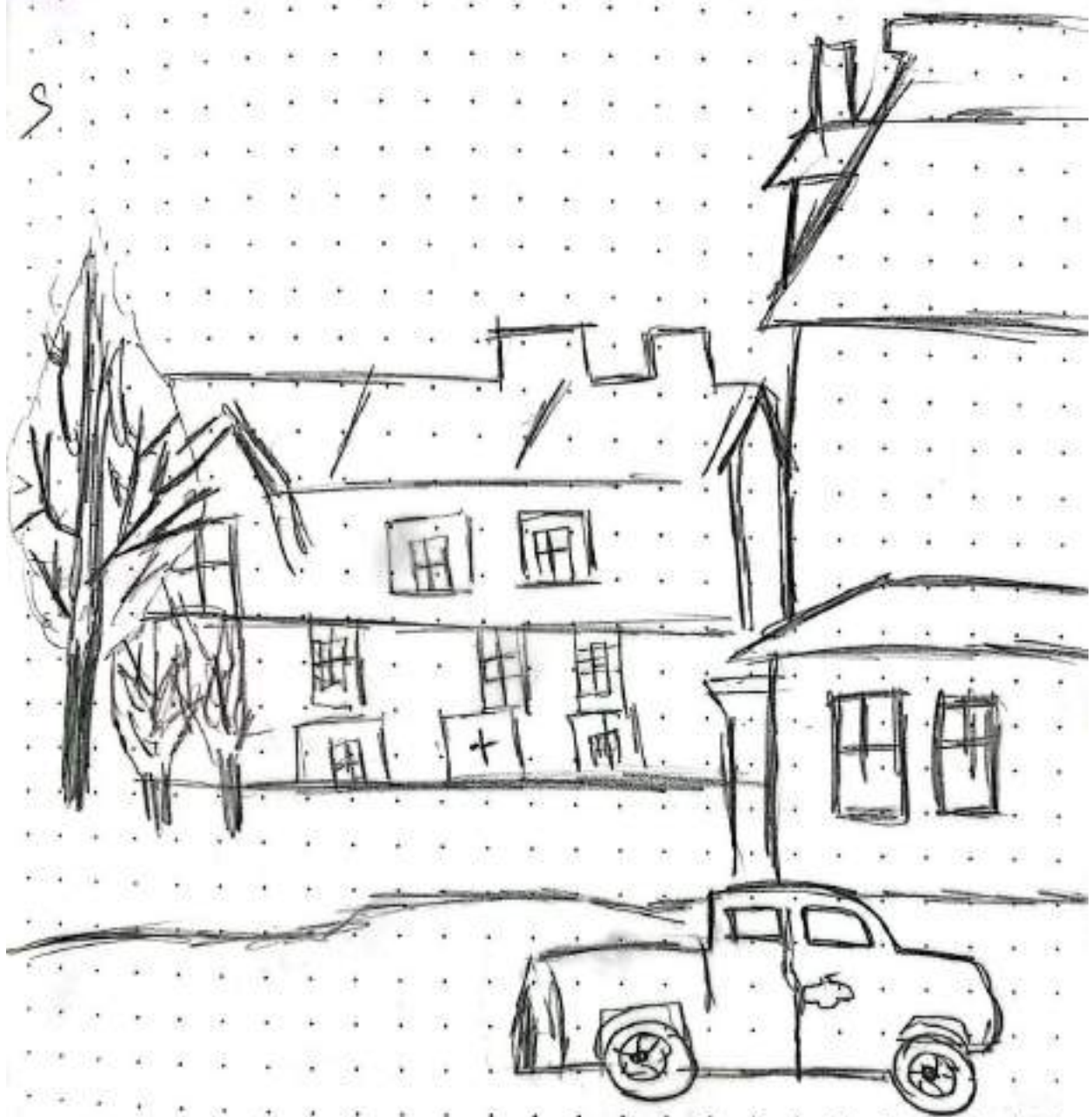


Major Religions

Delight



picture on the wall
(dorm building)



Bugs!

- The smell that comes from dorm faucet
- The water accumulated, after raining, on the pavement
- The girls in my floor who chat a lot loudly on the weekends

Project #3

Posters that help people question their beliefs/bias/stereotypes

Bring about social change

Seek Feedback!

↳ What do you think the poster is saying?

Ideas:

- 1 - Stereotypes about International Students?
- 2 - Perceived beliefs about Muslims in the U.S?
- Central Asia? → do people live in yurts? Communism? - stan countries
- Color blindness?
- Beliefs like Muslims are terrorists?
- ↳ history doesn't define current situation
- Veganism?
- "International Students are using up positions in the U.S"
- ↳ They take spots from American students
- Referring to Central Asians: "They're Russian," "It's all Born"

- "they stuck in Soviet mentality"
- - "9/11" → "All Muslims are terrorists"
- - "All - stan countries are instable"
- - Borat stereotypes
- - "Women are oppressed in - stan countries"

Feedback:

9/11 poster: moving the texts to the tower

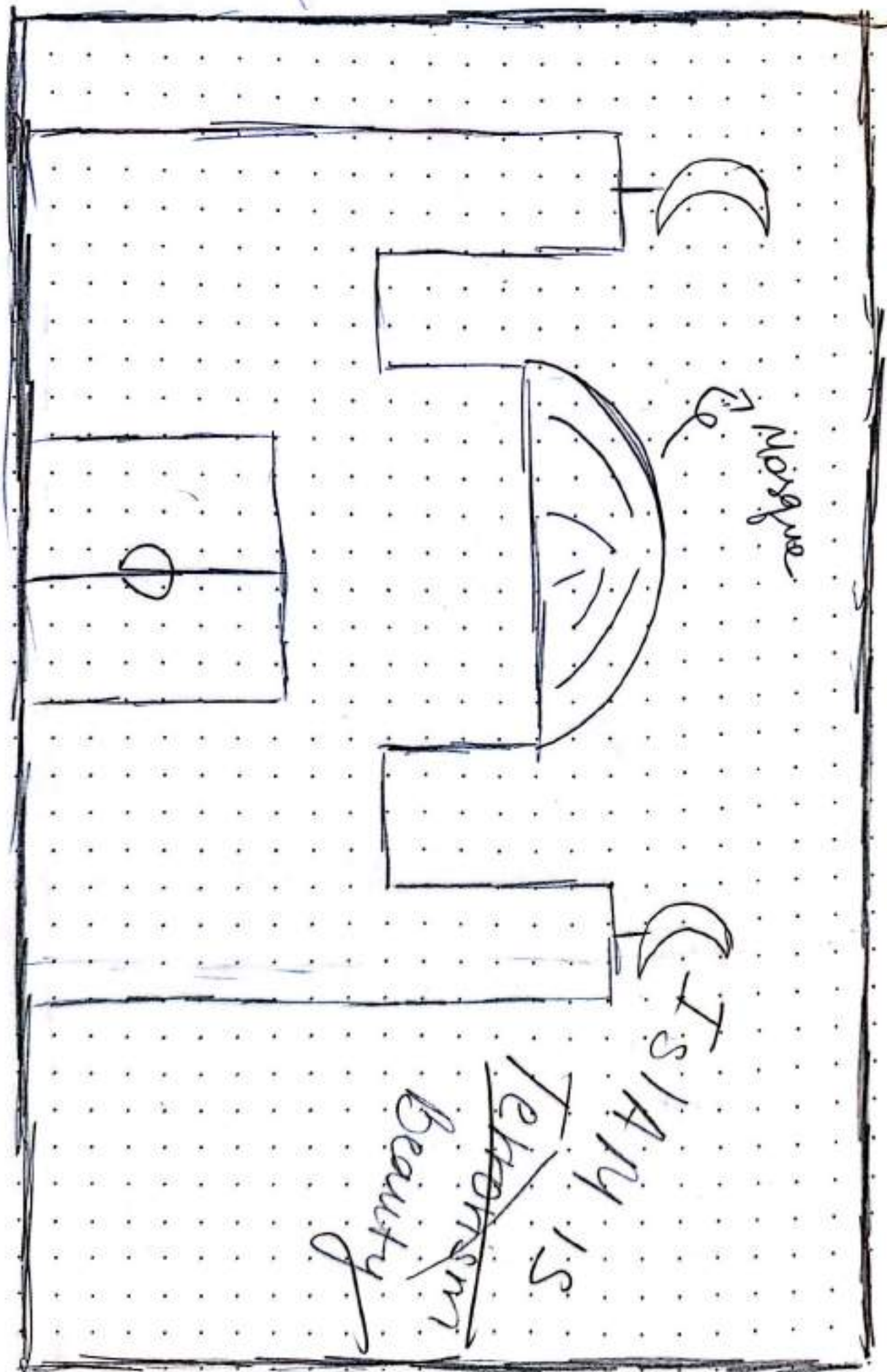
The third poster → mosque

[Three designs put into one place
Videos about Islamophobia? ✓

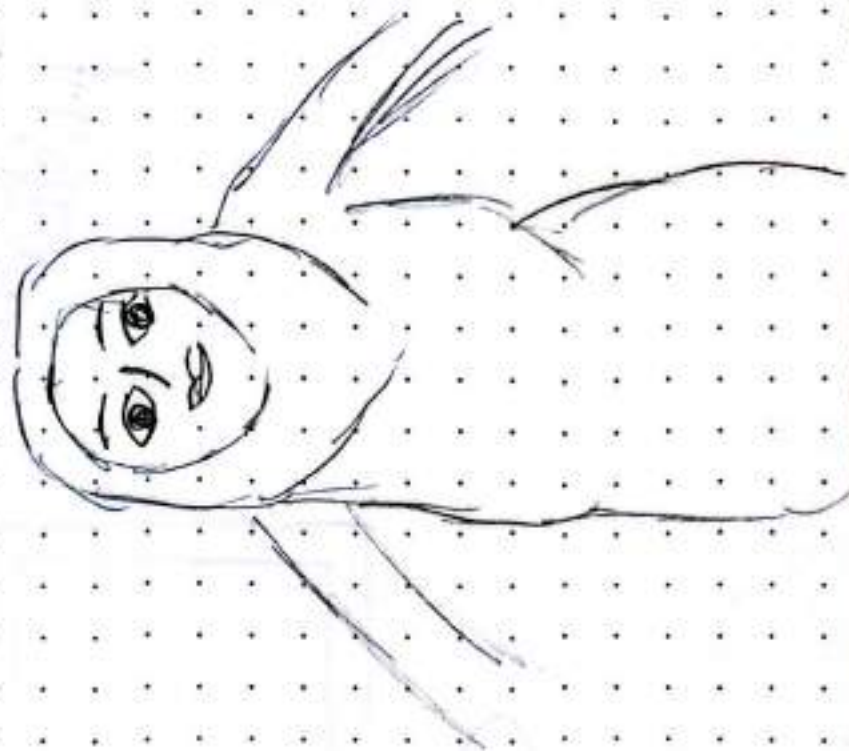
Is there any target?

Watch some Adobe Photoshop tutorials

- Ad campaign?



victim/
actor of
the belief

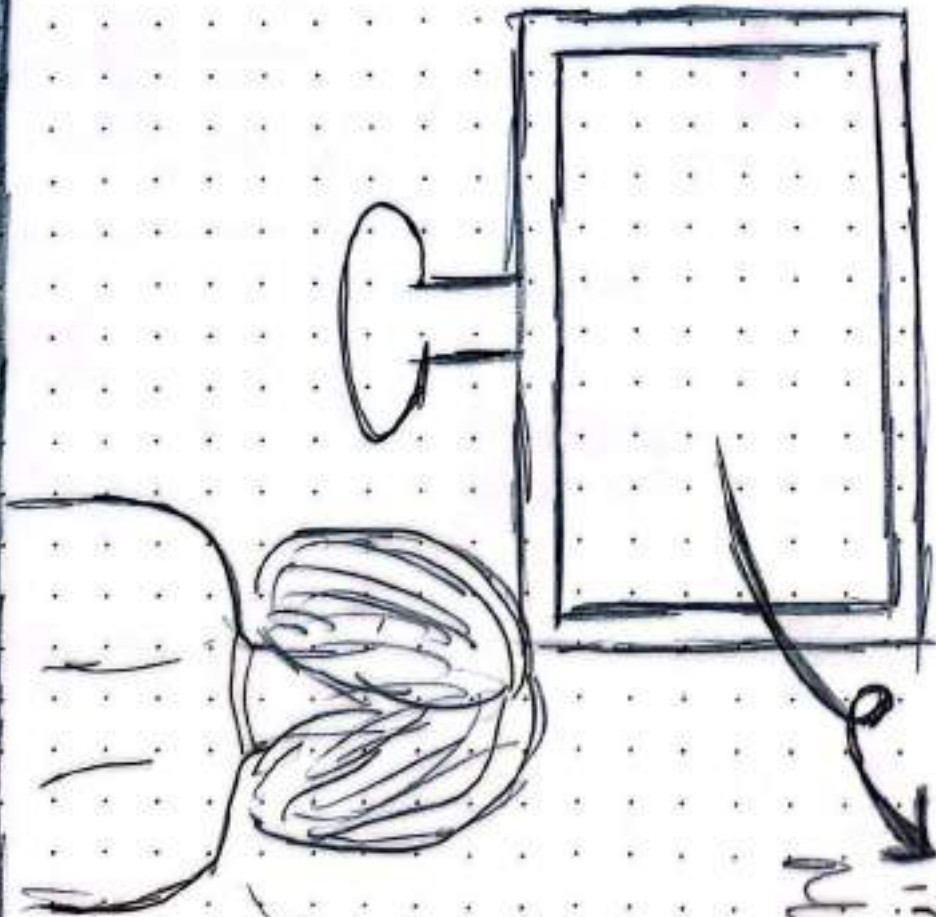


→ g/h

Terrorism has
no religion of

call to
action

Media
is Misleading



distorted
image of
Muslims

American
watching
misleading
news

Delights & Bugs

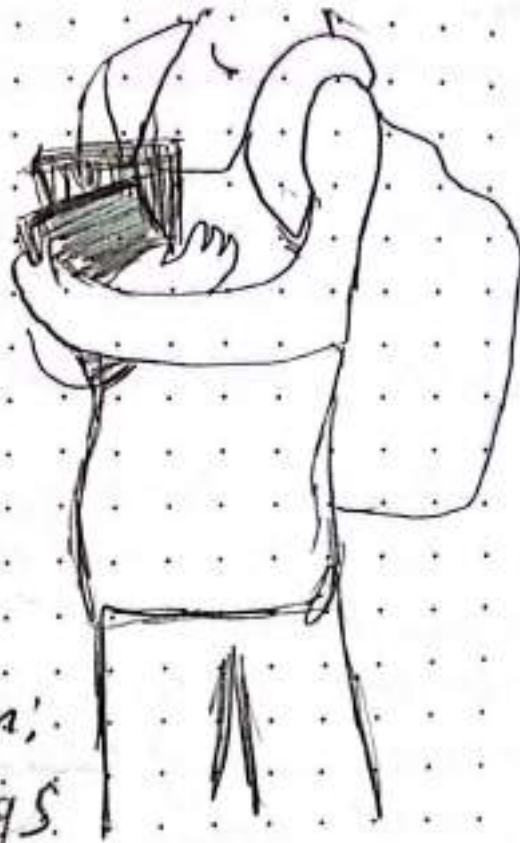
- It was sunny today. In a long time.
- It bothered me when I dropped my phone suddenly. It didn't break, fortunately.
- I'm taking a PE swimming class, I liked it!
- We hanged out w/ friends and played tennis together. It really made my day.
- The cozy and fresh and clean laundry right after washing them ~ such a delight!
- When walking with a friend, our walking pace matches perfectly!
- When a stranger holds the door for me.
- Woke up at 5 a.m! It's a delight that I can do what I set my mind to!
- When I was sitting on the Green, I liked the gentleness of the wind waving my hair.
- Realizing that New England spring is really beautiful and fresh.

Delights:

- Waking up and seeing the sun shining
- Taking long walks on the Pine Park
- It's warm outside and I can study on the Green!
- Eating iconic foco cookie before class
- Going to Foco and realizing it's only few people there
- That moment when a stranger puts effort to pronounce my name correctly :)
- Really and actually enjoying lectures in classes.

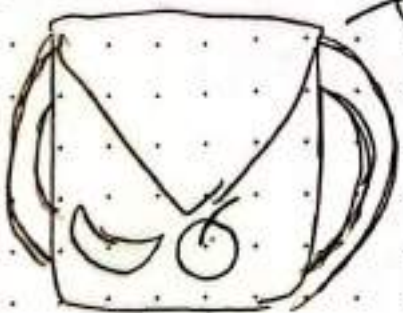
Bugs!

- Seeing my friend on the another corner of Green!
- Having to depend on the internet too much, even for homework
- Everyone is so busy...
- The fact that I'm lying on the Green and it feels like I should be grinding
- Having a lot on my plate in one day
- Forgetting to turn on alarm clocks "
- Running out of pencil in the middle of the class
- Having to balance jobs and classes
- The women's bathroom in my dorm is smelly



People on the Green;
usually biking w/ bags.
walking w/ bags





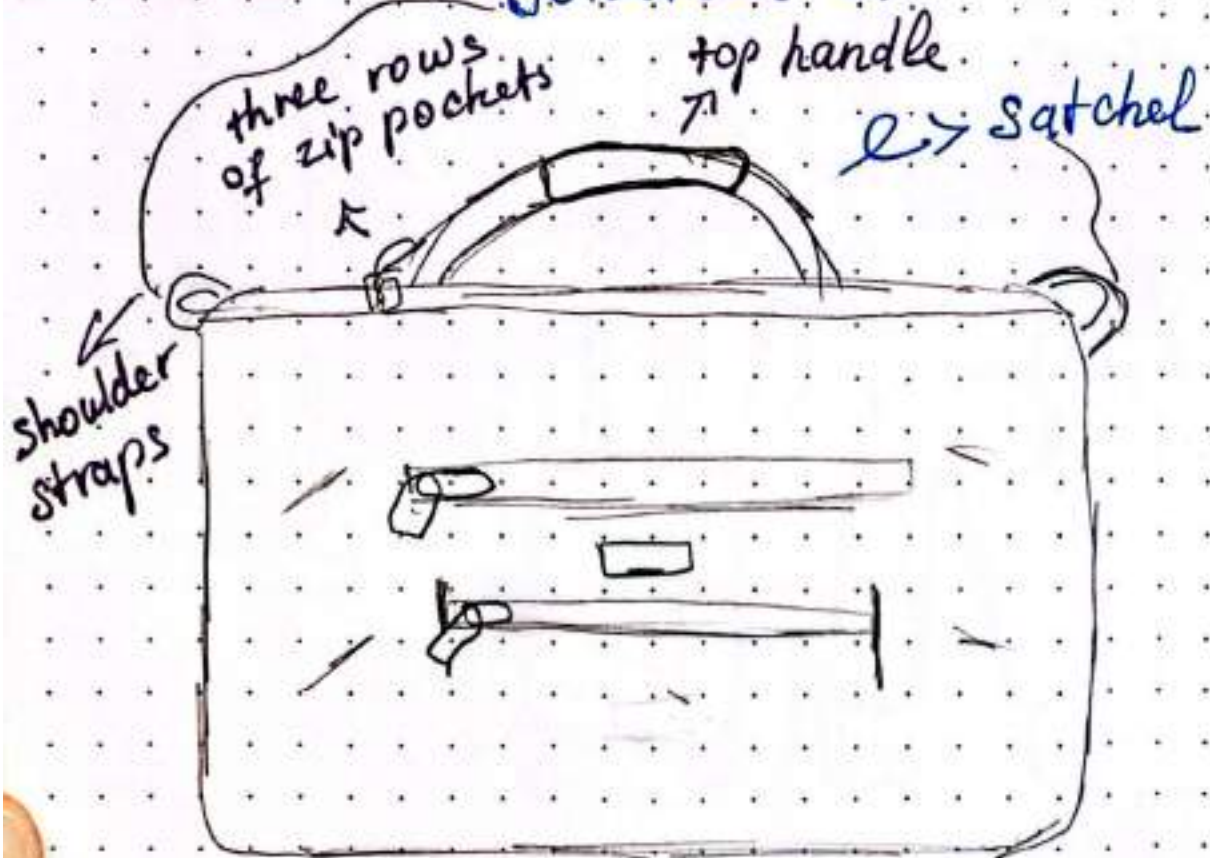
food goes bad
inside of backpacks



it's usually
cold in the mornings,
so students wear outside layers. Then, they
have to take off them and carry w/ them
everywhere



looks
stylish, but
hard to fit things



Frustrations: "I can't fit a lot into it"

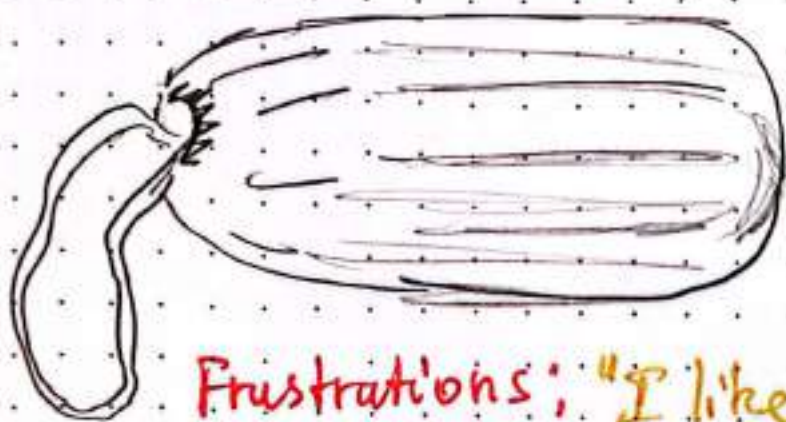
- There is no exterior pocket for a water bottle

- "Wish there was an in-built charger"

Story: "I bring my satchel everywhere: to classes, club meetings, etc. Unlike bag packs, I prefer this, because it doesn't hurt my shoulders. My dad bought for me."

Li'la G.

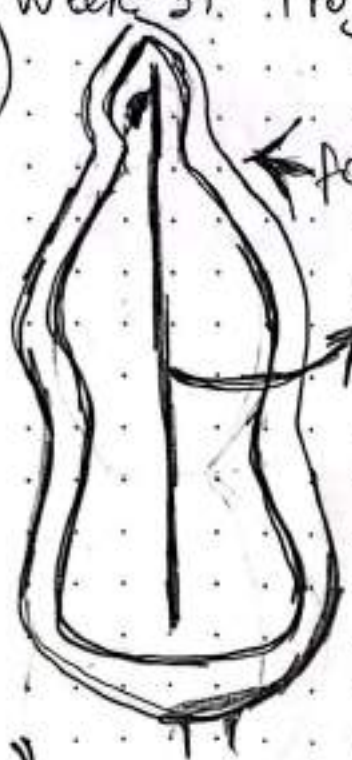
picnic
blanket



Frustrations: "I like lying down on the Green or on the Pine Park, but you see such picnic blankets need to be dense into that cloth material. What's more, I wish it had more pockets to put small items. And the handle also not user-friendly."

Story: "After classes, if I feel that it's a nice day to take a break outside, then I go to my room to pick up my blanket. I use it in a lot of locations."

1)

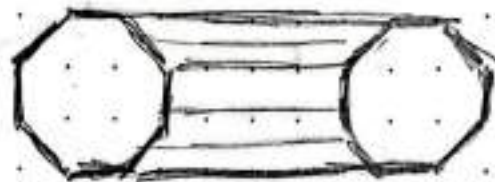


Cookie container

material (on sewing machine)

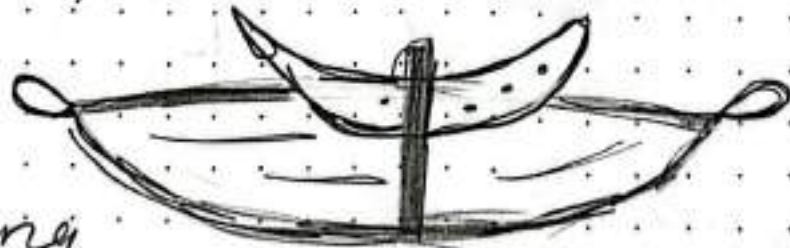
Let us introduce you
our loco cookie
container!

2)



Transportable backpack!

3)



Banana

Hammock

Features:

- attachable to the bag
- detachable for other things
- Can keep bananas for longer time as fresh.
- Don't get smushed inside.

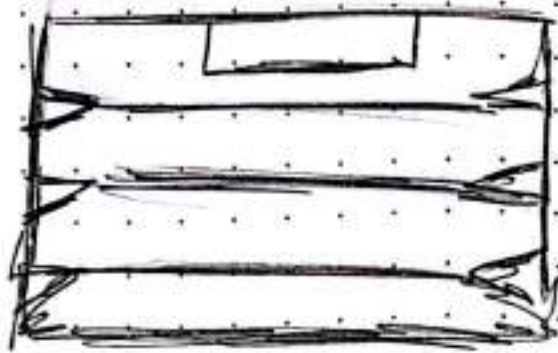
Delights!

- Stumbling upon a friend in the dining hall after a long time
- Seeing trees blossom!
- Finally finding a time to listen to my favorite song.
- Being able to go to sleep earlier
- Calling my parents in another continent
- I like when POCO serves what I like
- When a professor puts an effort to pronounce my name.
- Running in the forest alone

Bugs!

- when people walk so slow...
- I don't like when people harshly close doors
- I don't like when people aren't aware of their clothes in the washing machine.
- when I receive bad news consequently in a day
- When I'm very tired, have no time for hobbies
- Ants in my room!
- long lines for lunch
- when my favorite is damaged
- scam calls!

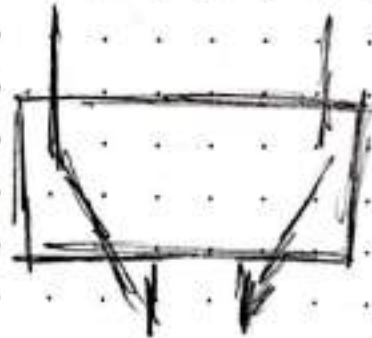
4)



Cookie Coochie

- adjustable
- pressable
- keeps cookies from breaking

5)



Helping people who use trochet to ease their meal process.

A lot of students struggle to pick their meals while their legs hurt or who are disabled.

Works-like vs. Even though it's not fully functional as long as people can get the idea, it's easier to move forward from there.

looks-like prototypes; it's better to use with designs (e.g. Photoshop) first before working w/ prototypes

14. Feedback:

I would like on the accessibility and visibility of the prototypes.

We might want to focus on the poco cookie container.

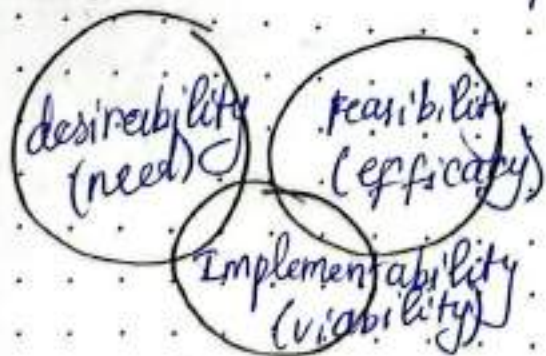
Some questions to consider:

- what are the feasibility?
- what assumptions do we have about
- = What's the main essential ^{the user?} feature?

Looks like - Works like - Interacts like

Testing $\neq$ Selling. Test known unknowns.
Test unknown unknowns.

Concept Evolution Map



We have a cognitive bias to focus on things we can most control, rather than things that are most important.

Presentation Tips:

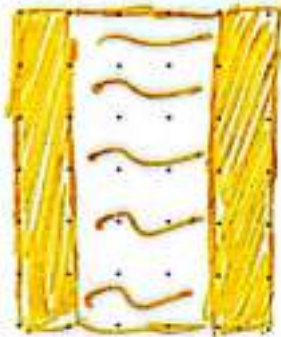
- Presenting is story telling.
- Bring along everyone w/ you.
- Tell the whole story.
- Start w/ the need.
- Use clear background.

Slide text is more ^{less} key phrases
line by line
Use photos



⇒ foam

material



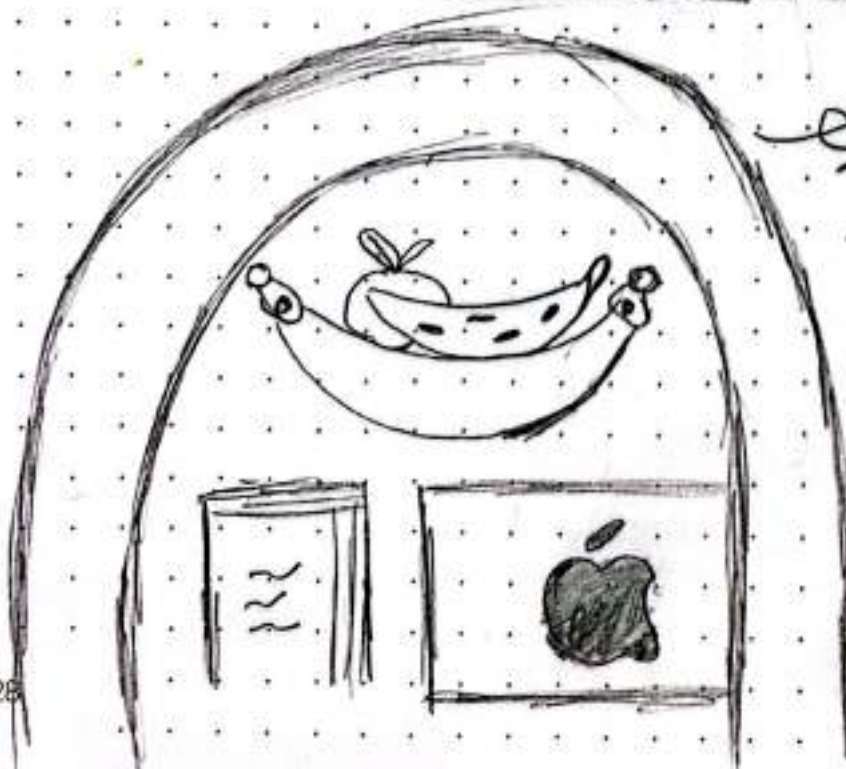
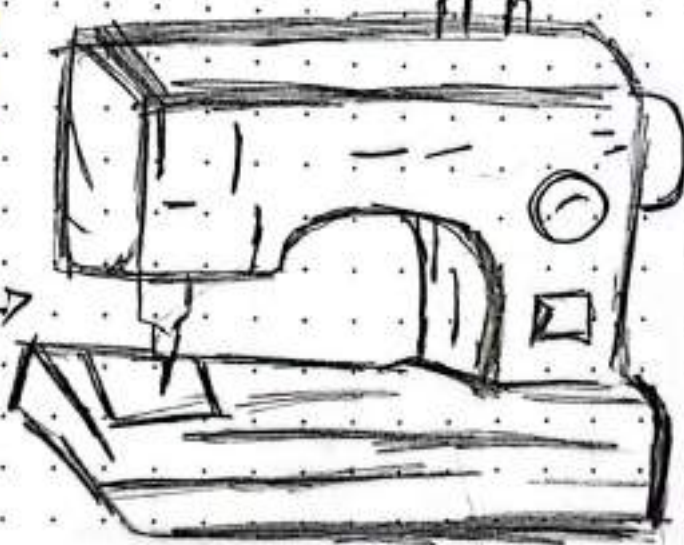
⇒ banana hammock (flat format)



⇒ dense format



sewing machine



⇒ how the prototype looks inside of the backpack

- Ideas:
- Education: VR experience for labs at schools? Safety needs?
 - Parenting: online child-care?
 - Digital (no physical)
 - Inventing VR pets for those children whose parents don't allow pets at home?
 - What are the ways to integrate technology into learning, parenting, recreation, healthcare etc. to imagine 2035



imagining Mars
and other planets for children
inside of a bus

could this be transformed to another
object/vehicle?

How do innovators like Elon Musk

imagine the future for children?

What are some existing projects/
interventions?

Who is the target audience &
beneficiaries?

Week 6!

- What are some ways to increase educational access in developing areas?

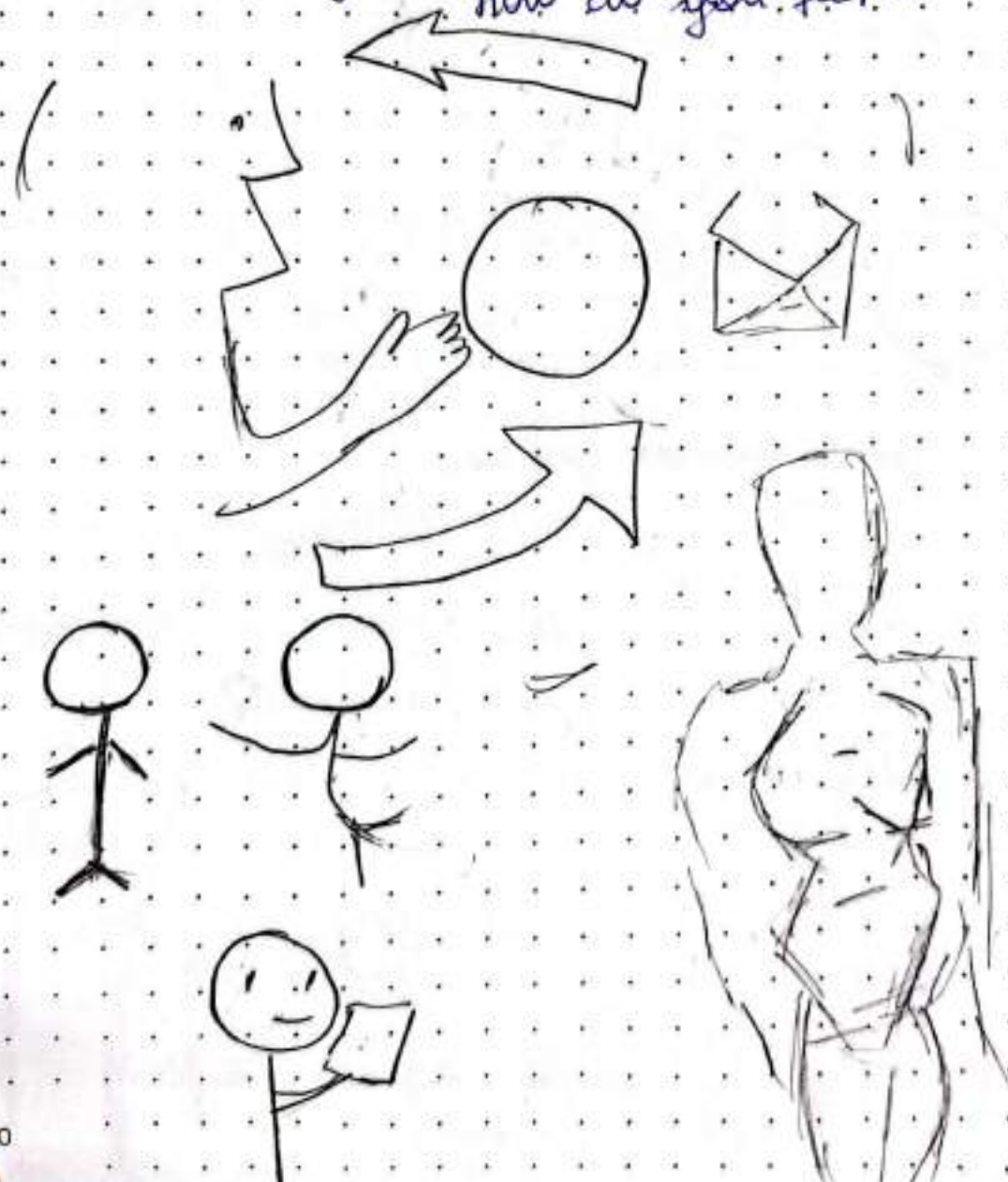
"How will — change from now to —?"

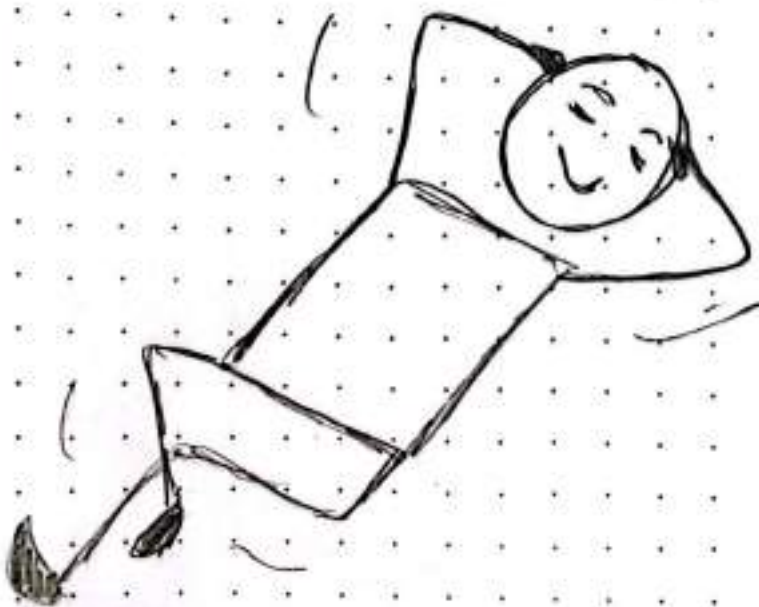
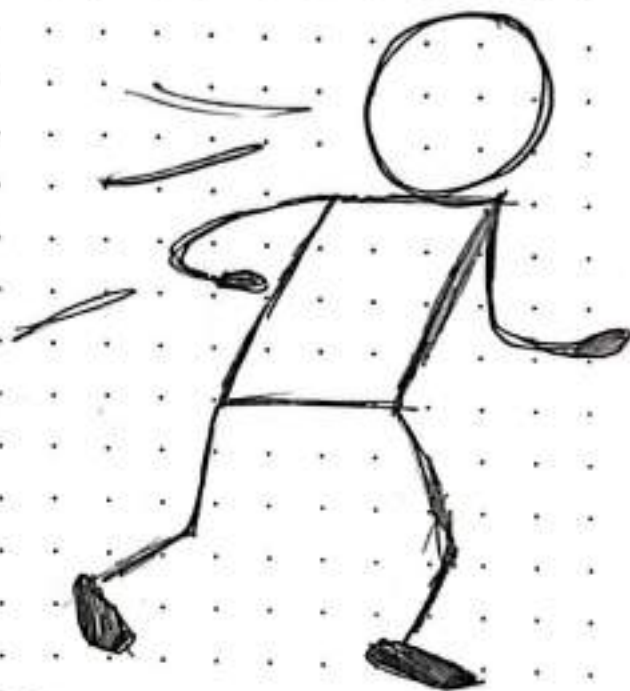
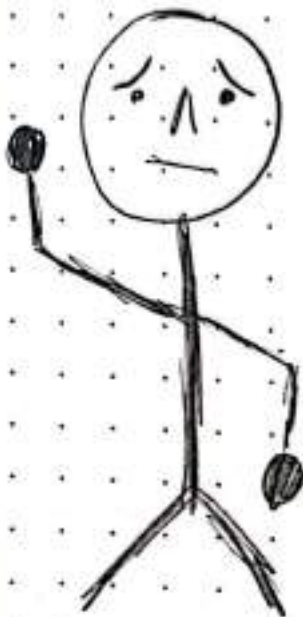
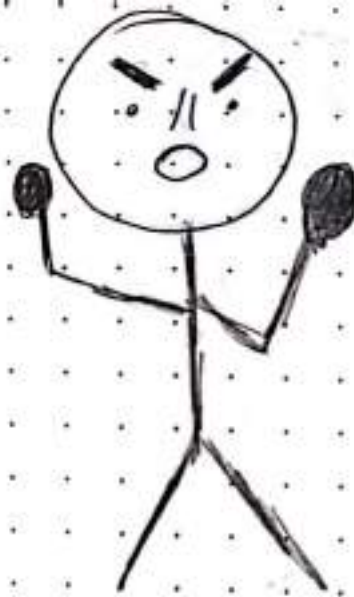
Needs: - Low-cost technology

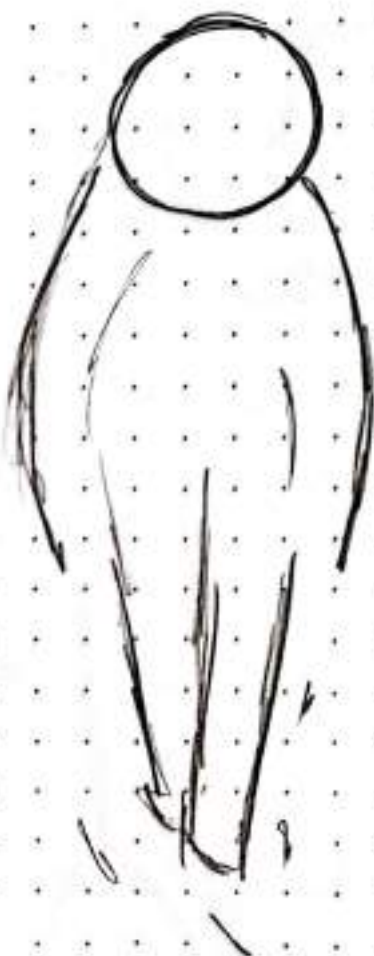
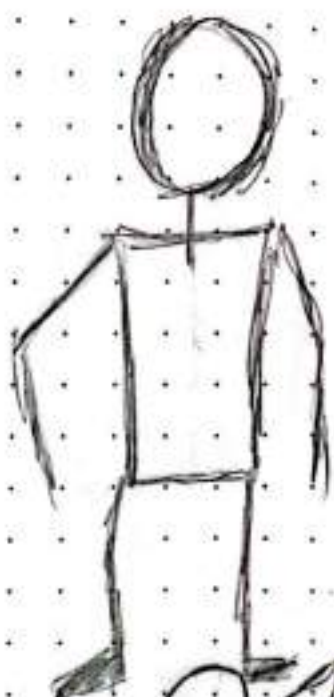
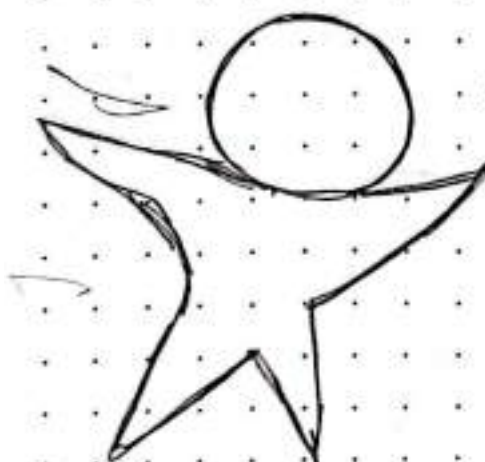
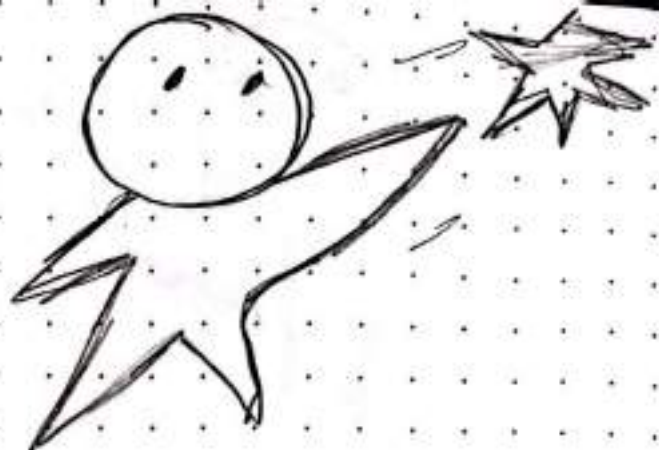
How can we imagine future where X can be applicable to many climates and places?

Cost is also an issue

Interaction design: How do you
How do you feel...



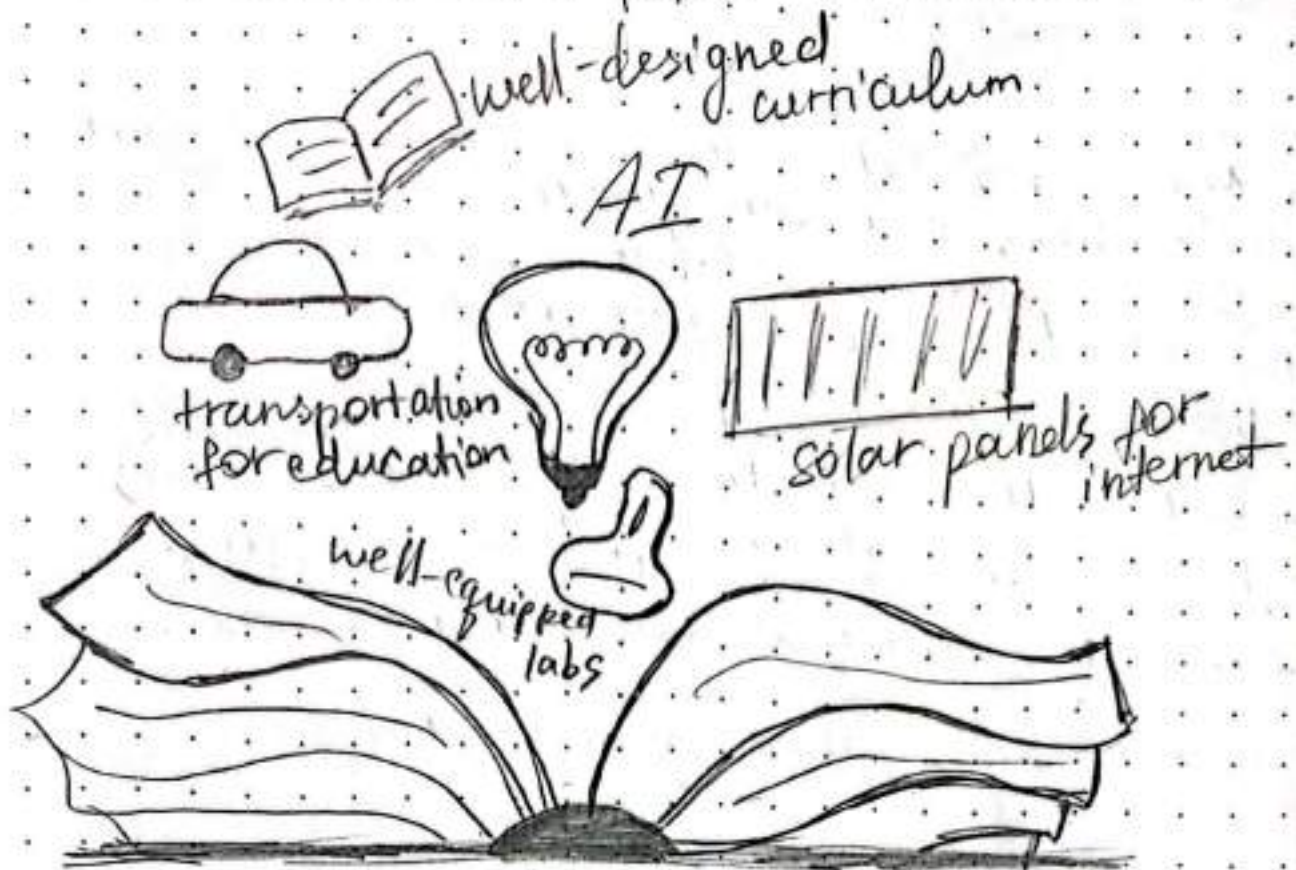




- [Bugs]**:-
- Gloomy days with lots of clouds
 - having a 10-hour difference w/ my parents.
 - Financially struggling...
 - My friend, "It's really annoying to wait in long lines in dining halls."
 - Being from a country where everyone needs an explanatory from me.
 - When a prof keeps misspelling my name.
 - Going to sleep late (past 1:30 am)
 - Not being able to wake up after two alarms
 - Having mid-life crises in a gloomy day
 - Being even more perplexed after the class

- [Delights]**:-
- Listening to a study playlist while reading a book
 - Living in a single room (It's really rare for many students, It allows me some downtime)
 - Seeing flowers everywhere on campus!
 - Saying hi loudly while riding a bike to a friend!
 - Going to bed so satisfied of what I've done in a day
 - Meeting with my favorite professor
 - Surviving six weeks of classes!
 - Celebrating my friend's birthday!
 - Realizing my own birthday is less than a month away!
 - Being able to keep my room tidy!
 - Seeing the tiny plant grow in my room!
 - When someone praises my writing ~
 - Having some leisure time to be lazy and lie down on the Green!

Education + Tech in 2032



[Interviews] Amen from Yemen:

- Because of war, girls are afraid to go to school (physical assault + risk)
 - traditional classroom setting
 - early marriage
 - limited access to internet and technology
 - limited resources for language learning
 - Dartmouth is really different
 - from Myanmar;
 - Civil war since 2011 → afraid to leave the U.S.
 - Sending girls to school is dangerous
 - refugee parents, especially, struggle to provide proper education for girls
 - Technology is good in cities, whereas rural schools have low resources
 - Dartmouth is automated
- Arshad from Afghanistan:
- Has five sisters → it was always challenging for girls even before Taliban
 - limited co-ed - Almost no internet
 - Dartmouth runs on internet

Joel from Sri Lanka:

- Private school (the British rule)
- his school was co-ed, but public school run on gender-based
- In the U.S., he doesn't see much difference because he went to a well-funded school
- The culture in Sri Lanka can limit girls' access to education → sports as well
- The pandemic changed the game

Vani from India: - from Delhi

- Girls living in cities have relatively more opportunities compared to those in rural areas
- Traditional roles restrict how girls exercise their rights to education
- Early marriage
- India in cities is like "West"
- Schools in cities have good access to tech and internet

Accessible education for girls in developing countries?

Questions: How can we imagine a "future" or "futures" (pluralism is better for creating possibilities) where girls in war-torn and environmentally-impacted areas can enhance their literacy skills? Where and how does tech and AI come into play in achieving this goal?

- What if education comes to them in areas where going to school is not feasible?

Problems:

- Traditional roles (e.g. having to do chores instead of going to school)
- War/Political upheaval (e.g. Afghanistan, Ukraine, Palestine, Yemen, Myanmar)
- Environmental (e.g. need to walk long distances in African countries)
- Family constraints (e.g. preferring boys' education over girls')
- Financial costs (e.g. education usually ends for girls after primary or secondary school because they can't afford higher education)

(Innovations + Statistics)

ENERGY MAP

Step 1

List your top energy-giving and energy-draining activities in which you engage regularly.

sleeping sitting on a chair too long
showering walking too much
talking to friends spending too much time on laptop
organizing a meetup (e.g. birthday) hearing bad news

Step 2

In the graph on the right, draw a bar of each one's (+) or (-) energy effect, as it shows up in your typical week.

Step 3

What do you notice about your energy patterns?

Dependent on a particular day
and how many assignments I have

Step 4

What are some doable, easy changes you can make to improve your energy flow?

I can work in chunks instead of overdoing assignments

+

+ -

+ -

TIME

Week 7!

- Bugs:**
- Sore legs after running
 - Unflushed toilet in the dorm's bathroom
 - Cold days in spring
 - Oversleeping a scheduled meeting
 - Promising to call my parents but forgetting it and they went to sleep....
 - Slow internet when logging in to see something important
 - Working on a design but forgetting to save
 - Long lines of to-do lists
 - Going to sleep past 1 a.m.
 - When I sign up for something fun, but I don't get chosen
 - Being a beginner at something
 - Forgetting an important notebook for class discussion

- Delights:**
- When a friend compliments my dress!
 - An early sunrise from my window
 - When a friend texts me that she misses me
 - Having the ability to smile
 - Calling a friend over two hours
 - A photo of myself turned out very cute!
 - Going for a stargazing night with a friend
 - Listening to one song over five times a day
 - Standing up for something some cause
 - Noticing the beauty of ^{T. cherry} spring and value
 - The ability to make independent choices and decisions as a female
 - Going to the Muppet Theater with friends
 - Being able to save money
 - When my presence is important for someone
 - When I win a gift card!
 - Writing a poem for a class
 - Filling a journal w/ thoughts

119 million girls are
out of school
due to conflict and
environmental crises

Message



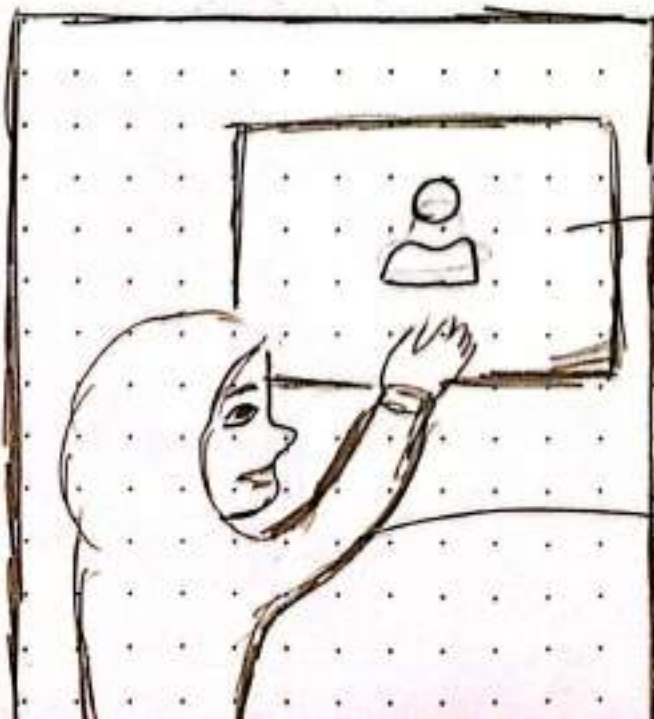
girl in
Afghanistan

Education in
Her hands

Reframe



Our
Solution



AI tutor
on the projects

At home
learning

By 2032, will smartphones still be dominant platform? or will it be something else?

Feedback: - what are some ways to explain with less words?

- "I think your idea is clear, maybe add several more posters?"
- The text seems to be dense.
- Is it something detachable maybe?
- How can we think about diversity and inclusivity in our device?
- What are the ways we can connect things of the current?
- What makes it different from a watch?
- Can a family member take it away?
- What are current interventions?

Notes for the final submission:

- Can be a video / Photoshop
- + slides (like the previous one)
- Project reflection
- Is it doable in 2032 or too "ambitious"?

Honest - User Feedback

- I like it's not phone or internet dependent.
- Can be accessible to rural areas.
- A.I. can go offline?
- What are some language limitations?
- Distribution might be problematic.
- Not apps

Next steps:

- other devices: jewelry?
↳ feels personal.
- Cost ↳ where can we get funding from?

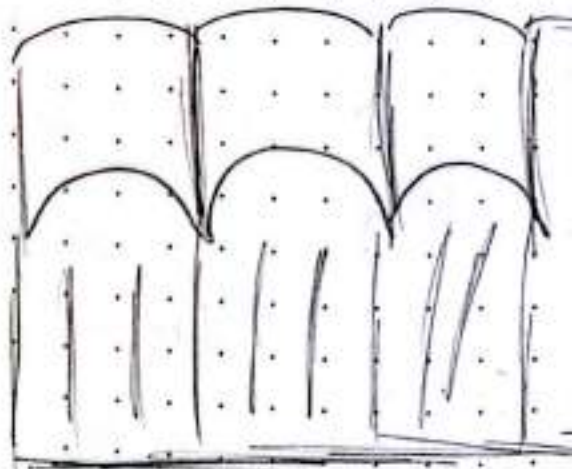
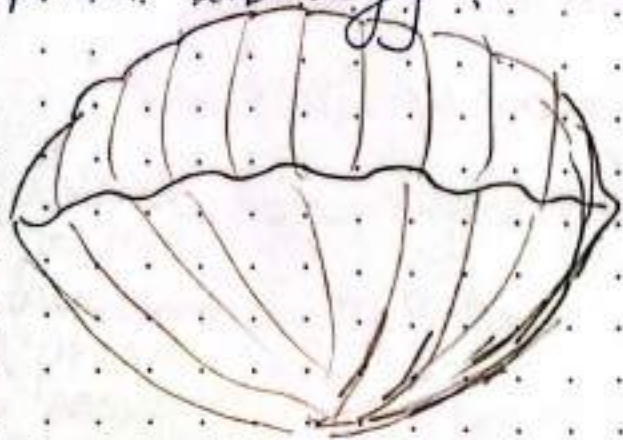
Slide notes:

- What's the opposite?

- What's the inverse?

Thinking with analogies.

Direct analogy:



Personal Analogy:

"I would imagine myself as an immune system engaged in combating a virus."

Compressed Conflict:

2-word analogy bringing different things into unified connection.

- Involves creating a paradox or contradiction by conceiving of two opposing ideas as being currently true.

Compelling need + cool innovation.

Storytelling; concise; point of view; challenges assumptions; brings ideas to life; feels authentic; shows doesn't tell; believable; focuses on people.

Functionality isn't a story. People are the story.

Designing your life I:

- Everyone says that the college 4 years will be the best years of your life.
- Each generation is told that "lazy, entitled, going nowhere / won't grow up / just lost"
- "If you did things right, you would've everything settled once and for all by twenty-five"
- "the miserable process of getting from 25 to 35..."
- "when you're scared stuff"
- "... it repeats..."
- Energy can be like GPS for life: helping you figure out what to steer towards, and what to steer away from.
- Noticing what gives you energy and what drains your energy
- Awareness is a prerequisite for empathy
- You can empathize for yourself and those around you.

Emotional help people see the importance of an idea

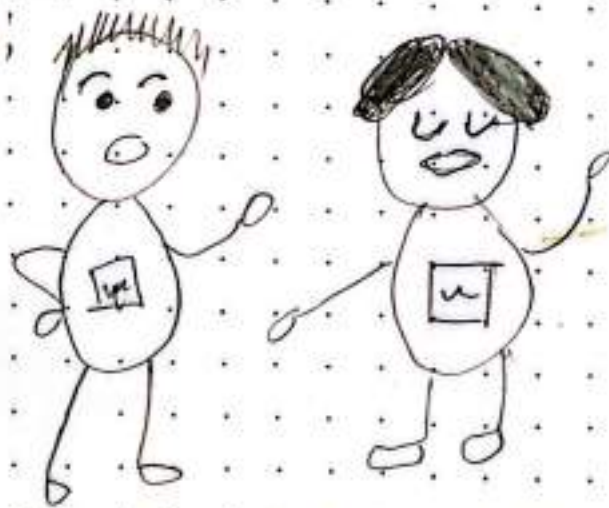
Stories: empower people to use an idea through narrative

Things to avoid:

- focusing only on technology
- focusing only on convenience

Slide Notes:

Fun Activity!



Janus: Roman god of beginnings and endings, transitions, gates, and doorways, time, and duality.
Janusian Thinking: "Compressed Conflict" is a phrase (usually just two words) that captures a paradox or tension in a design problem. By focusing on resolving a specific tension, our design solutions become both precise (useful) and non-obvious (innovative).

Three forms of feedback:

- Appreciation - "see, acknowledge, connect, motivate, thank" - Valued
- Coaching - "help expand knowledge, sharpen skill, improve capability" - Seen
- Evaluation - "rate or rank against a set of standards, to align expectations, to inform decision making" - Stability

Core design skills: soliciting and noticing feedback

Success:

S-Simple: find the core of any idea

U-unexpected: grab people's attention by surprising them

C-concrete: make sure an idea can be grasped and remembered later

Credible: give an idea believability.

Week 8!

- Bugs:
- So much trash left after Greenkey
 - Putting lots of things on my plate, but not liking half of them.
 - Too many commitments.
 - "Wasted food at POCO" Tina
 - Loud music from Greek houses
 - Not working faucet
 - Silence during interactions with professors
 - "I have a lab that's midnight tonight"
 - "My Wisp Symposium"
 - Unusually cold today

- Delights:
- Having very good night's sleep
 - Being thoughtful about my decisions.
 - Green leaves
 - Meeting deadlines on time
 - Calling family in Uzbekistan despite time difference
 - Having a diverse community at Dartmouth
 - Greenkey!
 - Getting 10 out of 10 for homework
 - "We got some cool results from my physics lab" Bobby
 - "I negotiated some tough conversation"
 - "I'm finally done with the presentation" ^{Jack} Clay
 - "Proving my friend's point against Clay"
 - "I won in the video game" ^{Tina} Tina
 - The construction the Green finished
 - "Waking up" Donathan

How Might We? :- provide a clearer definition of "brave space"?

- Influence positively while being not divided?
- Collaborate with other universities?
- Pre and post encampment activities?
- Negotiate with Dartmouth Security?
- enhance Dartmouth Dialogue
- Create something that doesn't depend on financial aid?
- Create a direct communication between Beilock and students?
- Bridge the gap between classroom and news on campus?
- Make engage students around important topics while also handling intensive academic terms?



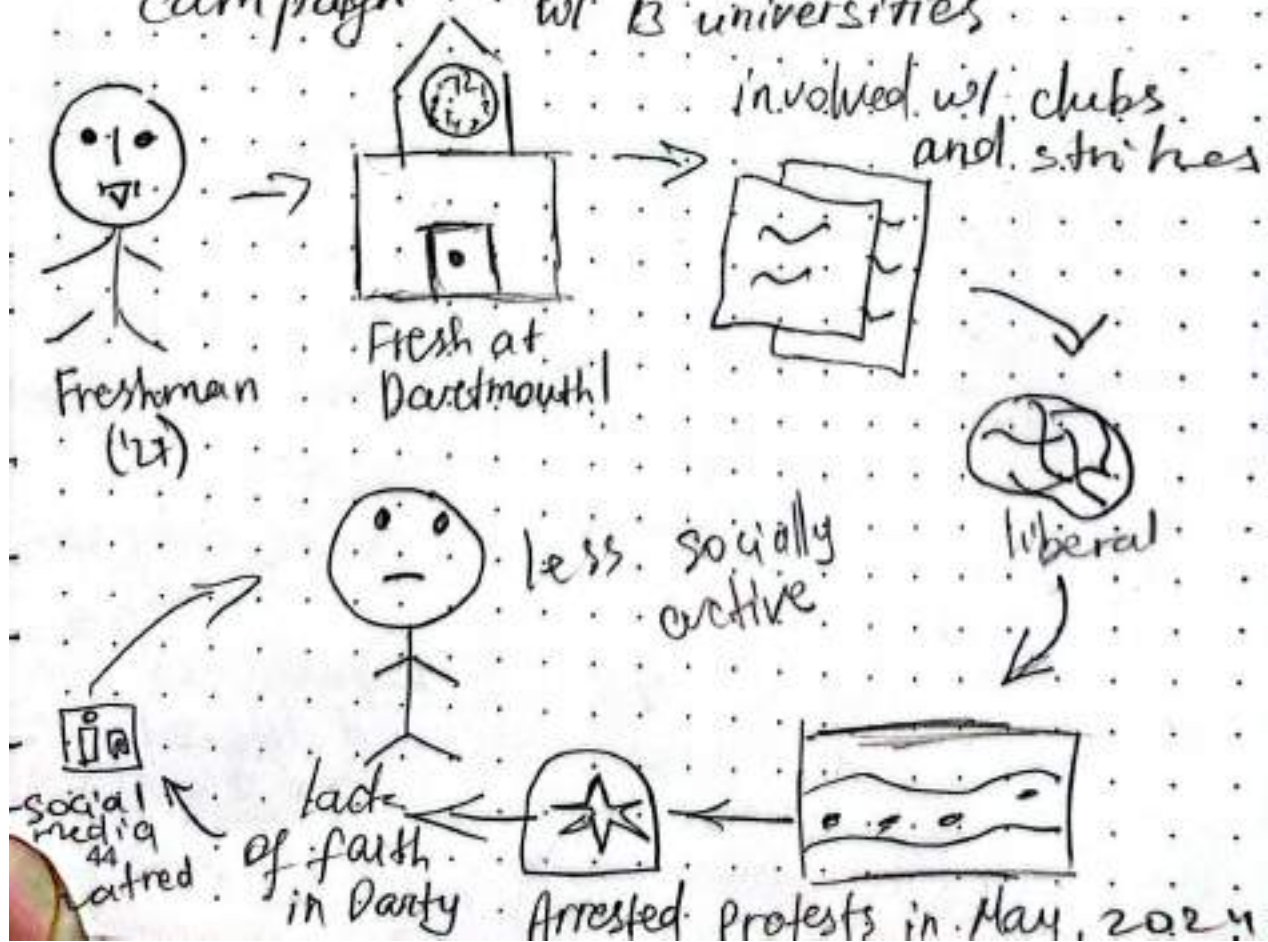
Freedom of expression at Dartmouth:

Ideas: - A website - AI-integrated app

- online petition collector?
- Ways to escape social/ideological bubble
- Students want to be anonymous

Secondary Research:

- There are a lot of student protests around American colleges
- At Dartmouth: Dartmouth Dialogue Project
- At Harvard: "Free Speech Guidelines" Various summits and conferences
- At Duke: "Friends for Duke" initiative
- NGOs, Gates Foundation, Obama foundation
- "The Campus Call for Free Expression" campaign w/ 13 universities



Designing Your Life II: Getting unstuck

Design Mindsets ; - Embrace experimentation

(iterate your way to an answer)

- Radical collaboration (work w/ others)

- Bias to Action

- Mindfulness of process

We all get stuck from big to small things

→ Final research paper topic

- Major at Dartmouth - what to buy for parent as presents when I come back - Final presentation

- what to draw as an inspiration

Try smarter, not harder. → concrete → specific.
→ complete-able

→ Taking a walk around Occom

Go to office hours w/ professor

Brainstorming a lot of ideas (idea parking lot)

→ Talk to a friend about the research idea

→ I will make a Google meet w/ them and verbalize my ideas

Design mindsets can be useful lens to help you get unstuck

when you're stuck, instead of "trying harder," see if you can adopt designers' mindset and "think differently."

Ask yourself, "what would a designer do?" ⁴⁵

Major-career linkage!

• Dartmouth Bubble → a lot of options to choose from

Social media marketing → psychology

Retail → economics / psychology / studio art

Human rights → public policy / history

Teacher → English / mathematics

Directly related / loosely related

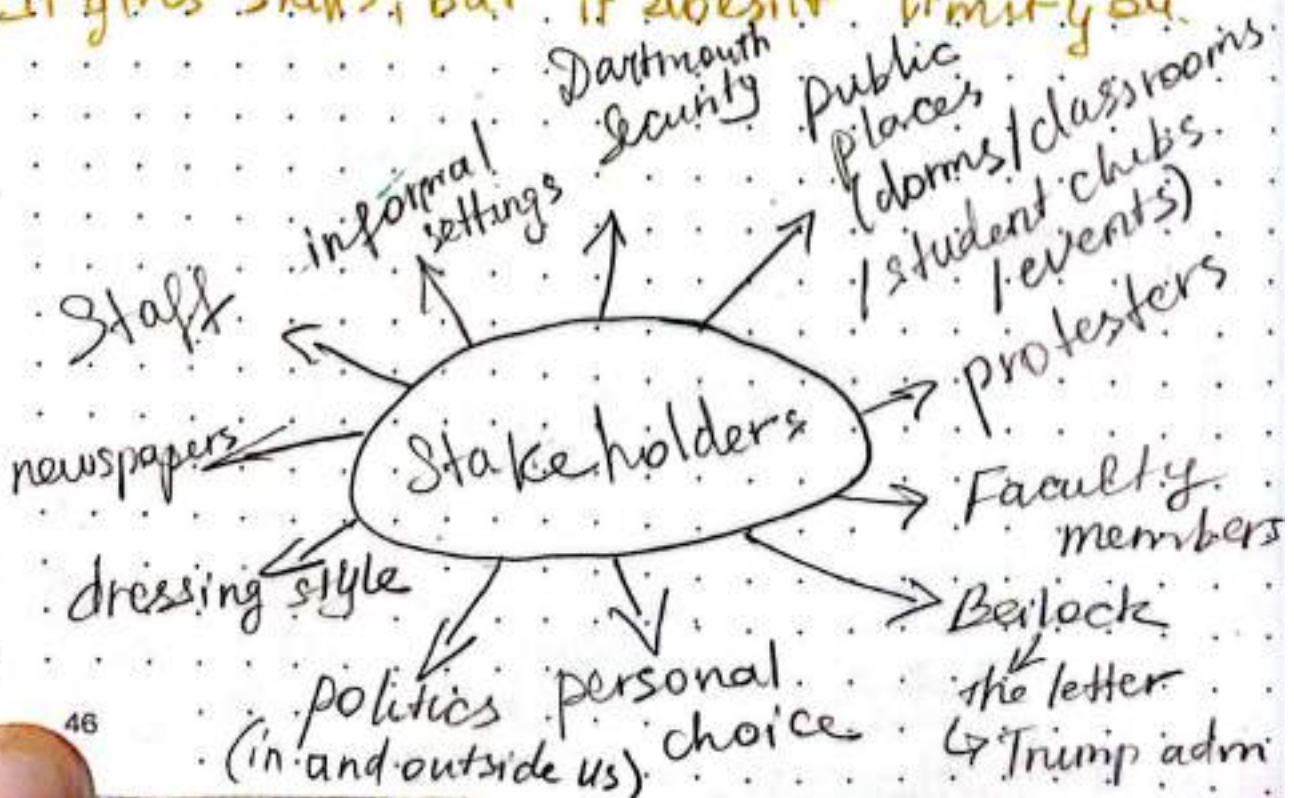
What are your actual interests and skills?

There are music majors who are working in Investment banking.

Don't get caught up in the question, "what are you going to do with that?"

Your major isn't unimportant

It gives skills, but it doesn't limit you.



Week 9!

- Bugs:**
- Loud noise in the early morning
 - Having to endure only on 5-hour sleep
 - Long classes and meetings
 - I need to travel 2 days to come back home
 - Needing to pay taxes
 - Mosquitos!
 - Accidentally messing up my room
 - Being internet-dependent all the time
 - The unpredictability of the future
 - When things don't go according to my plans and intentions
 - Long hours of travel spent

- Delights:**
- Sunny day!
 - Meeting at least four friends in the dining hall
 - Witnessing the change of seasons
 - Coming back home in two weeks!
 - Living in a single room on campus
 - Buying presents for family
 - Finishing an essay in one sitting
 - Painting, reading and finding some time for hobbies
 - Less time spent on the screen
 - When a professor really wants me to succeed
 - The limitless possibilities of majors, minors, etc!
 - Seeing people happy on the Green
 - Meeting new people
 - A bunch of love

"Dartmouth App"

- Direct communication between administration and students
- Can chat w/ Dartmouth Security
- Read news from other universities / online chat?
- Dartmouth Dialogue meeting times / schedule
- Pre and post encampment / protests news / collaborations
- know your rights → all in one place

Ideas:

Could it

be not

app-dependent?

→ wouldn't it

be internet

- dependent?

"Buzz"

- Connected to Canvas
- Space for opinions for students (mainly), professors, administrators
- Option of staying anonymous
- Sharing opinions in politics and controversial topics.
- Upvote and downvote

→ Wouldn't it

be a simple

solution for

such an

important issue?

Feedback:

Could Dartmouth

control Canvas?

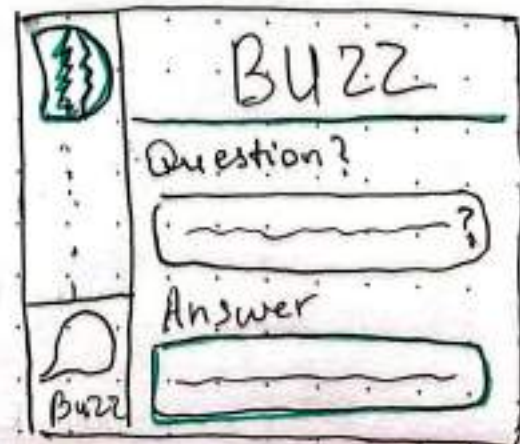
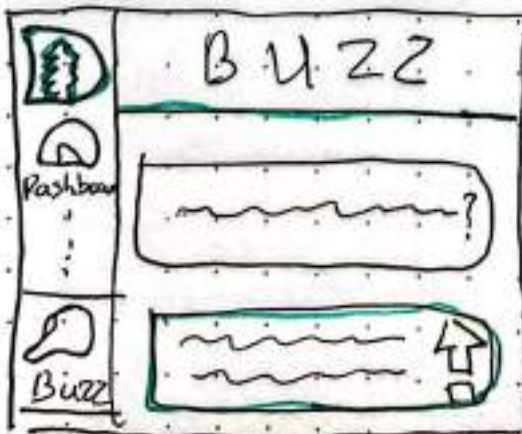
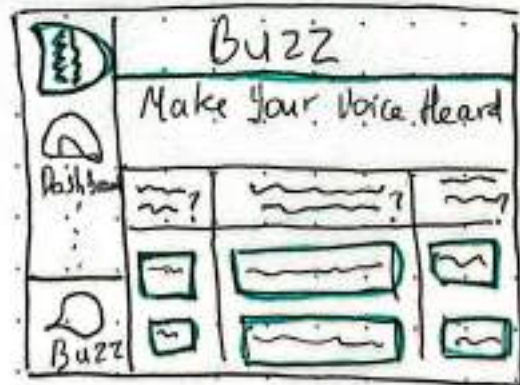
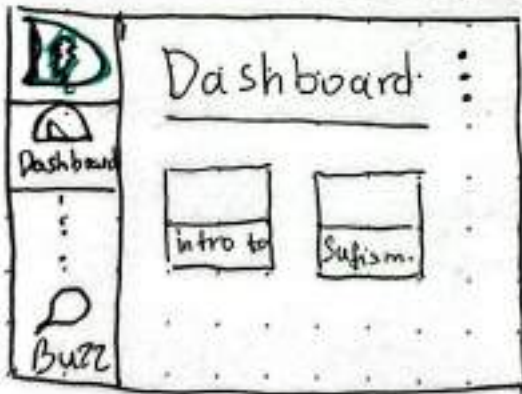
Wouldn't it interfere w/ studies?

Who controls it? Administrators or students?

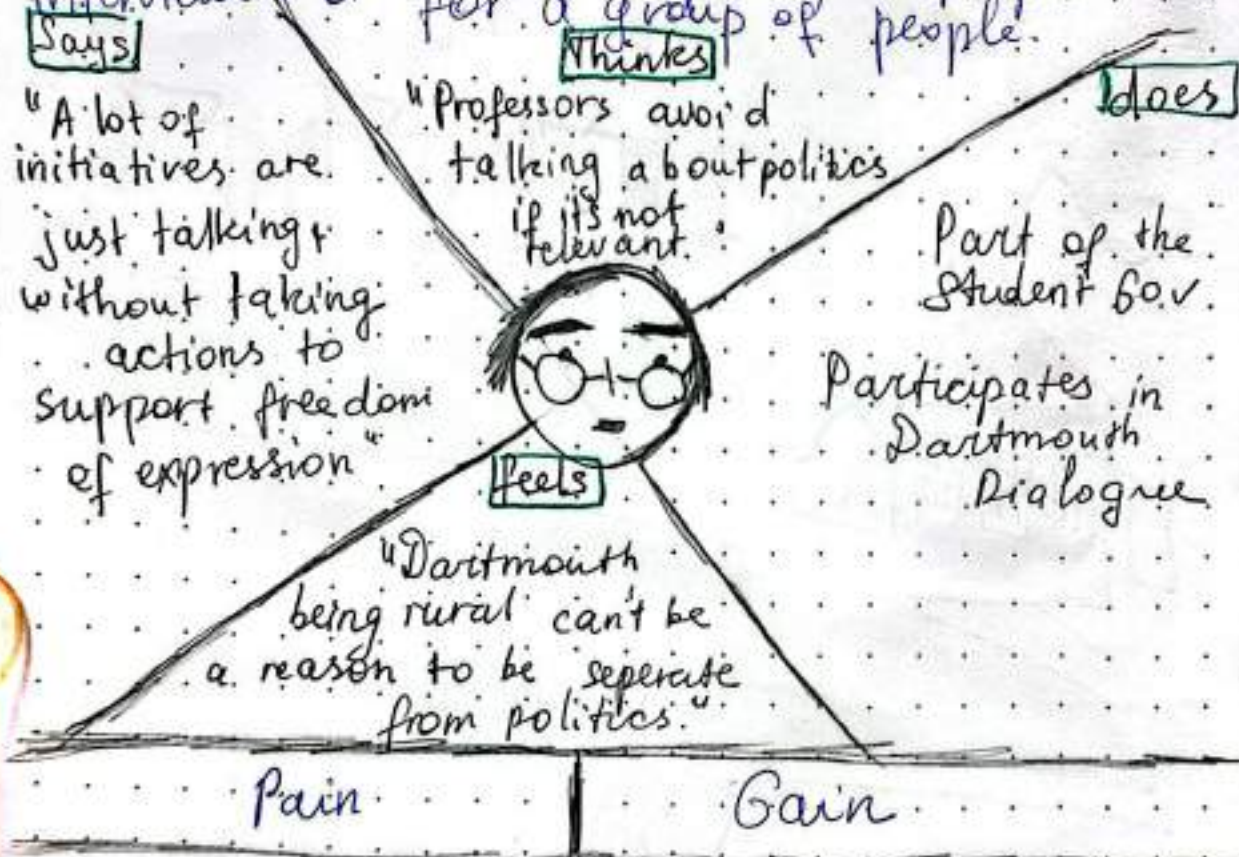
How much attention could it get?

120

What are pros and cons?



Notes: Empathy Map - a way to organize interview & observation data from a user's perspective. Helpful in finding patterns and contradictions. Use one empathy map per interviewee or for a group of people.



Positionality - practice of a researcher delineating their own position in relation to the study. It's the context of your identity, in terms of race, sexuality, origin, etc.

Ingroup / Out-group? Do I belong to this group? Is it Greek house? Am I unaffiliated?

Default assumptions & preferences. What are some generalizations am I making? What do I prefer?

Power dynamics. When I interview potential users? How might my attempted solution affect them?

Imagine a future that works. Be optimistic. We are bound by consensus. Conventional thinking. Learn to gain confidence when doubted by others. Breaking habits of fear.

and the illusion of control by others.

1. Stop saying "I'm sorry" for a week

2. Don't answer your phone (5x)

3. When someone saying something insulting, speak out...

What if your challenge is to change a system?

↳ Solving the components

Based on feedback today, create a defined version of the prototype.

Testing # Selling

Ecosystem/Stakeholder Maps → a way to visualize and understand relationships among the relevant constituencies.

Picture cards, creative toolkits, Participatory Design, Buzz reports

Keep a running list of assumptions you're making about your users, their needs and their situation.

Positive deviance (Bright spots)

Personas are composite portraits designed to represent a group of people. They are drawn from quotes, patterns, needs and observations and can be fictional.

Building an analogy:

Start w/ your situation →

Describe it in general terms →

Think of what else fits that description →

Enumerate some of its qualities →

See how you incorporate those back in →

B = M · A · P
behavior = motivation · ability · prompt

Think of examples.

A designed experience/object that doesn't work because the only missing element is Motivation → Ability & Prompt

↳ A designed experience/object that works well because all three elements are present.

Issue of Bias:

You should never accept the problem as given.

Instead, reframe the problem based on your observations and analysis of key goals.

Abstraction lets you understand the core of the problem.